Exp No:9

Experiment to understand Matplotlib library use cases in Data Science through visualization

AIM:

Use Iris data set to understand the Matplot lib library

PROGRAM:

```
import pandas as pd
import numpy as np
import seaborn as sns
import matplotlib.pyplot as plt
%matplotlib inline
data=pd.read_csv('/content/iris.csv')
data
data.info()
data.describe()
data.value_counts('species')
sns.countplot(x='species',data=data,)
plt.show()
dummies=pd.get_dummies(data.species)
FinalDataset=pd.concat([pd.get_dummies(data.species),data.iloc[:,[0,1,2,3]
]],axis=1)
FinalDataset.head()
sns.scatterplot(x='sepal_length',y='sepal_width',hue='species',data=data)
sns.scatterplot(x='petal_length',y='petal_width',hue='species',data=data,)
sns.pairplot(data,hue='species',height=3);
plt.show()
sns.FacetGrid(data,hue='species',height=5).map(sns.histplot,'petal_length'
).add_legend()
plt.show()
sns.FacetGrid(data,hue='species',height=5).map(sns.histplot,'petal_width')
.add_legend();
plt.show();
sns.FacetGrid(data,hue='species',height=5).map(sns.histplot,'sepal_length'
).add_legend();
plt.show();
sns.FacetGrid(data,hue='species',height=5).map(sns.histplot,'sepal_width')
.add_legend();
```

```
plt.show();
```

OUTPUT:

	sepal_length	sepal_width	petal_length	petal_width	species
0	5.1	3.5	1.4	0.2	setosa
1	4.9	3.0	1.4	0.2	setosa
2	4.7	3.2	1.3	0.2	setosa
3	4.6	3.1	1.5	0.2	setosa
4	5.0	3.6	1.4	0.2	setosa
...	...	...	...	...	...
145	6.7	3.0	5.2	2.3	virginica
146	6.3	2.5	5.0	1.9	virginica
147	6.5	3.0	5.2	2.0	virginica
148	6.2	3.4	5.4	2.3	virginica
149	5.9	3.0	5.1	1.8	virginica

150 rows × 5 columns

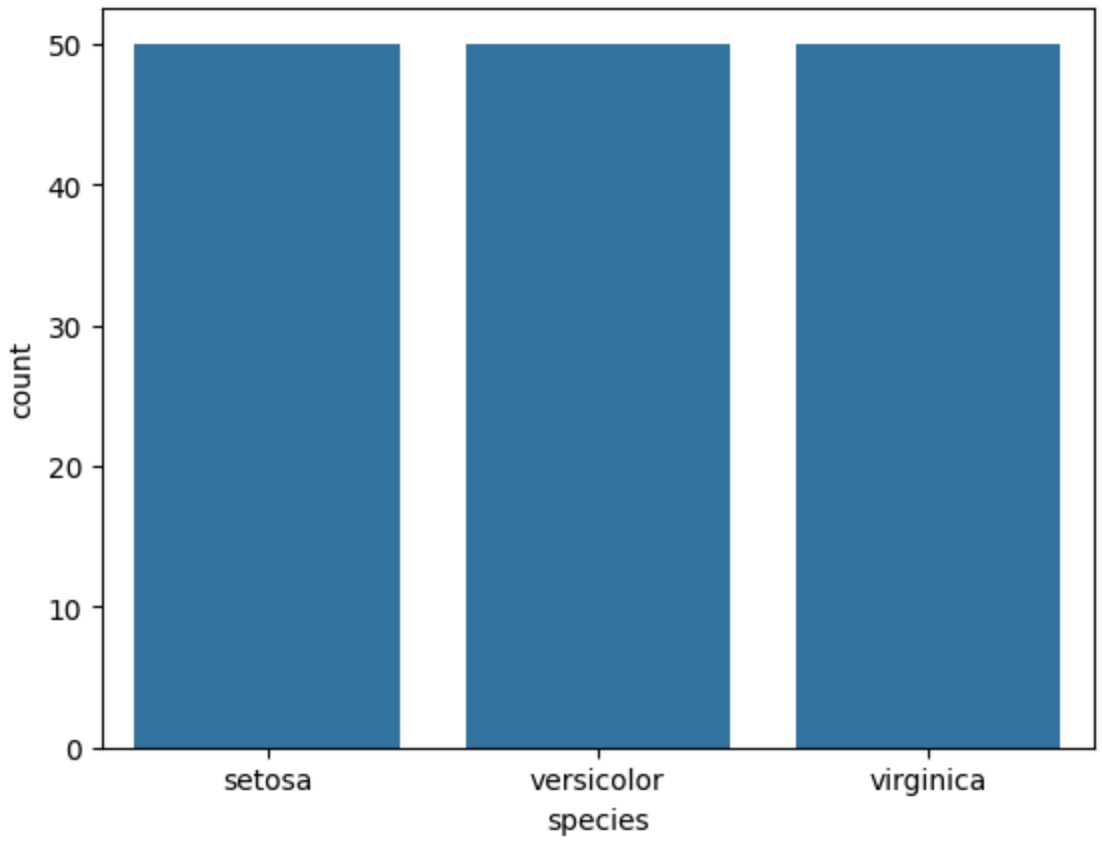
```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 150 entries, 0 to 149
Data columns (total 5 columns):
 #   Column        Non-Null Count  Dtype  
---  ------        --------------  -----  
 0   sepal_length  150 non-null    float64
 1   sepal_width   150 non-null    float64
 2   petal_length  150 non-null    float64
 3   petal_width   150 non-null    float64
```

```
 4   species       150 non-null    object
dtypes: float64(4), object(1)
memory usage: 6.0+ KB
```

	sepal_length	sepal_width	petal_length	petal_width
count	150.000000	150.000000	150.000000	150.000000
mean	5.843333	3.054000	3.758667	1.198667
std	0.828066	0.433594	1.764420	0.763161
min	4.300000	2.000000	1.000000	0.100000
25%	5.100000	2.800000	1.600000	0.300000
50%	5.800000	3.000000	4.350000	1.300000
75%	6.400000	3.300000	5.100000	1.800000
max	7.900000	4.400000	6.900000	2.500000

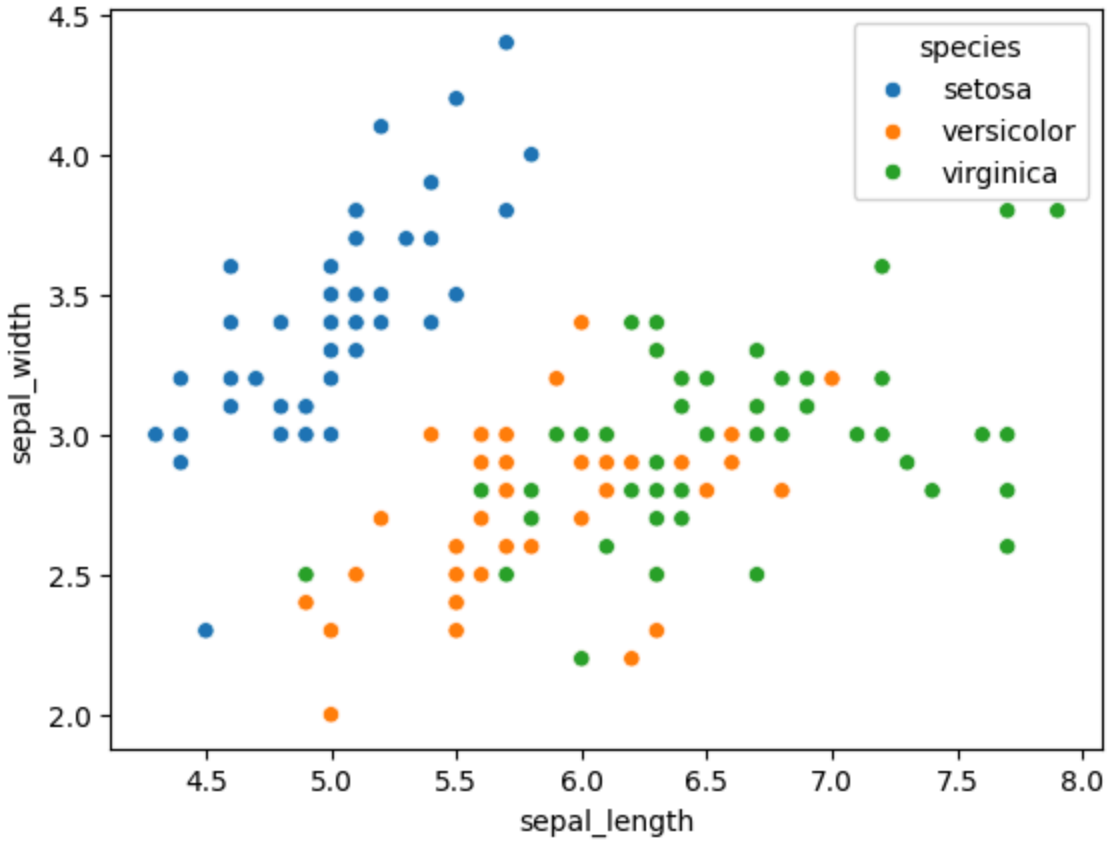
	count
species	
setosa	50
versicolor	50
virginica	50

dtype: int64

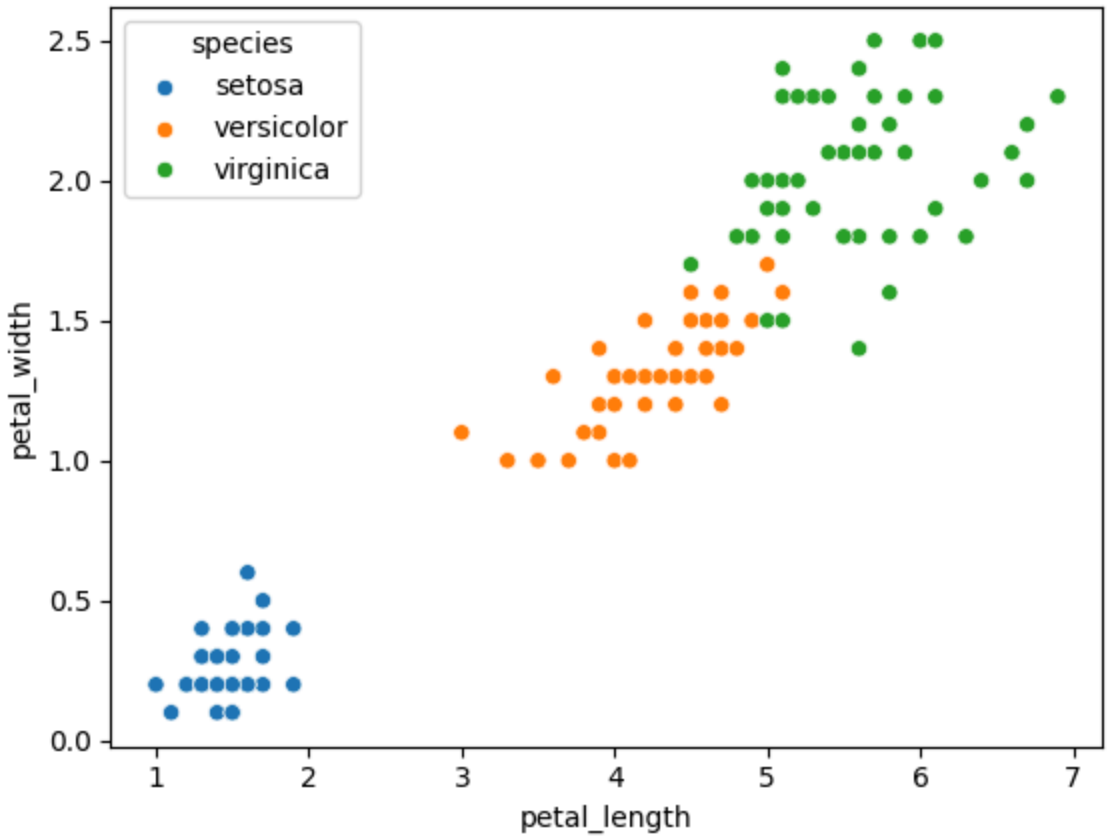


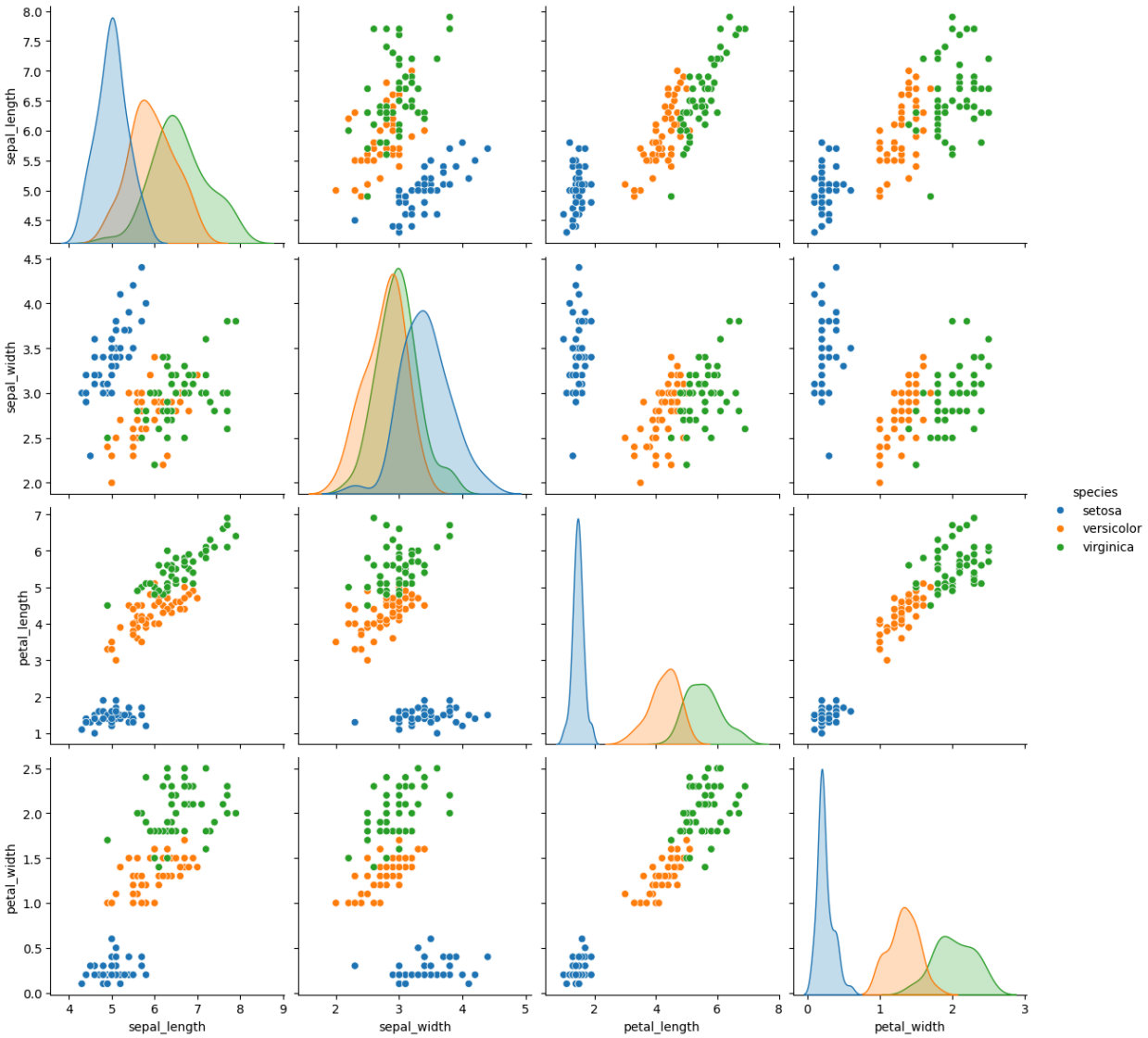
	setosa	versicolor	virginica	sepal_length	sepal_width	petal_length	petal_width
0	True	False	False	5.1	3.5	1.4	0.2
1	True	False	False	4.9	3.0	1.4	0.2
2	True	False	False	4.7	3.2	1.3	0.2
3	True	False	False	4.6	3.1	1.5	0.2
4	True	False	False	5.0	3.6	1.4	0.2

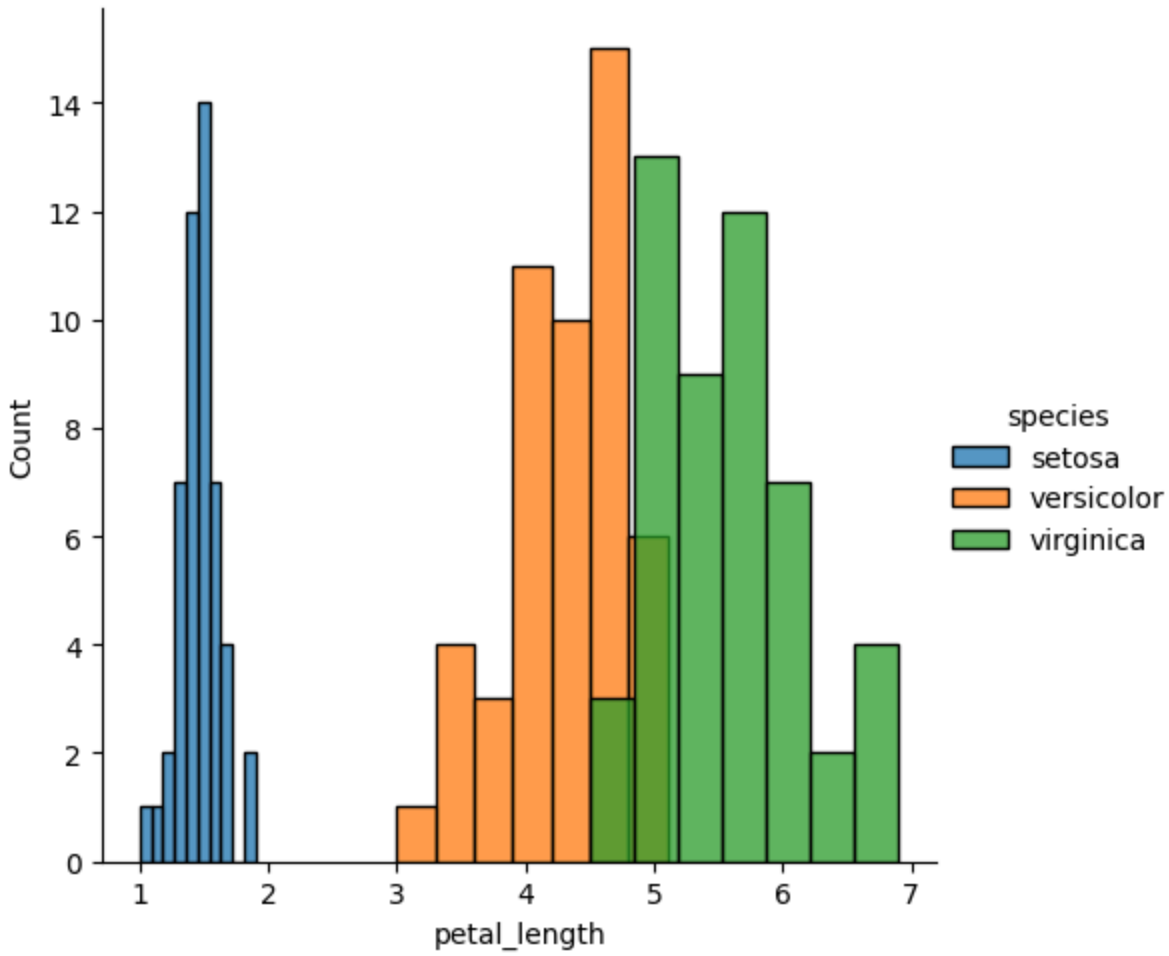
<Axes: xlabel='sepal_length', ylabel='sepal_width'>

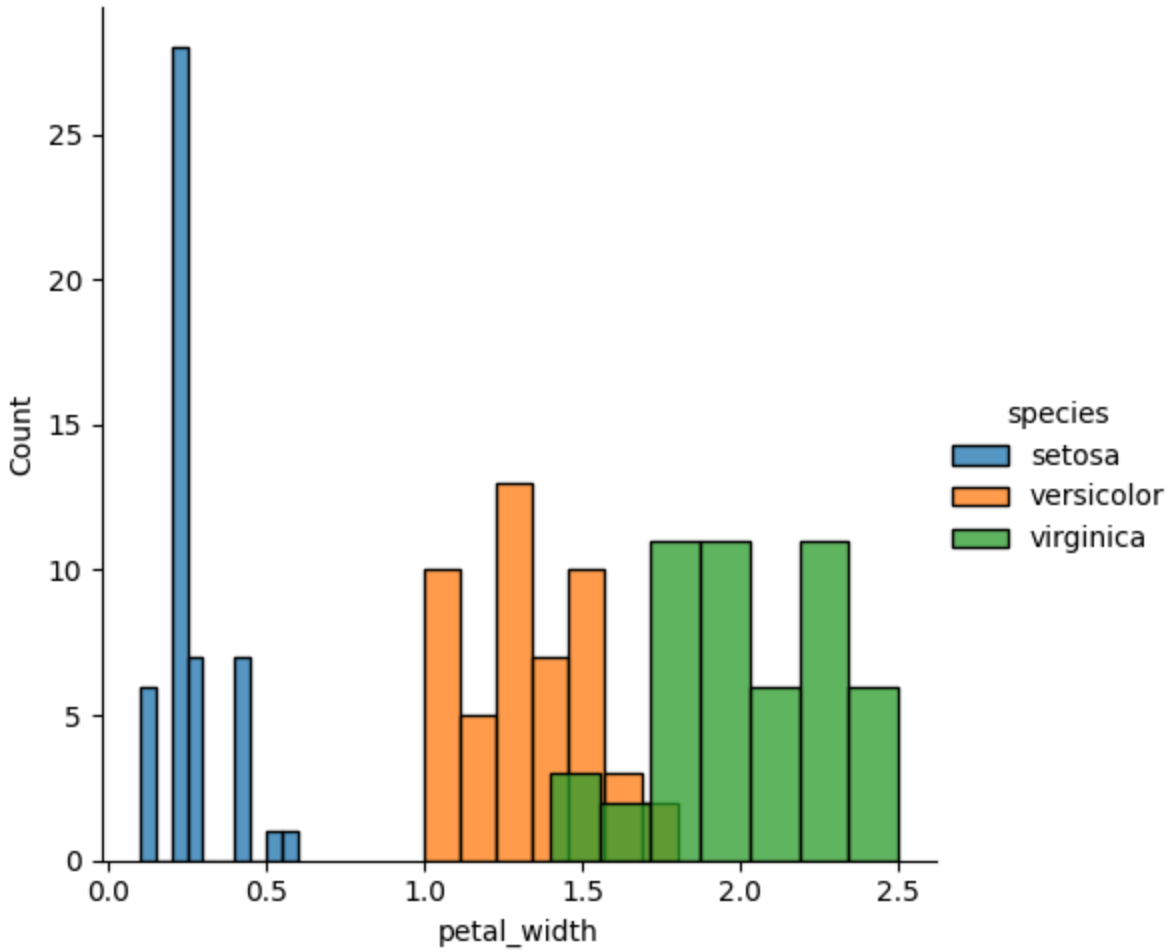


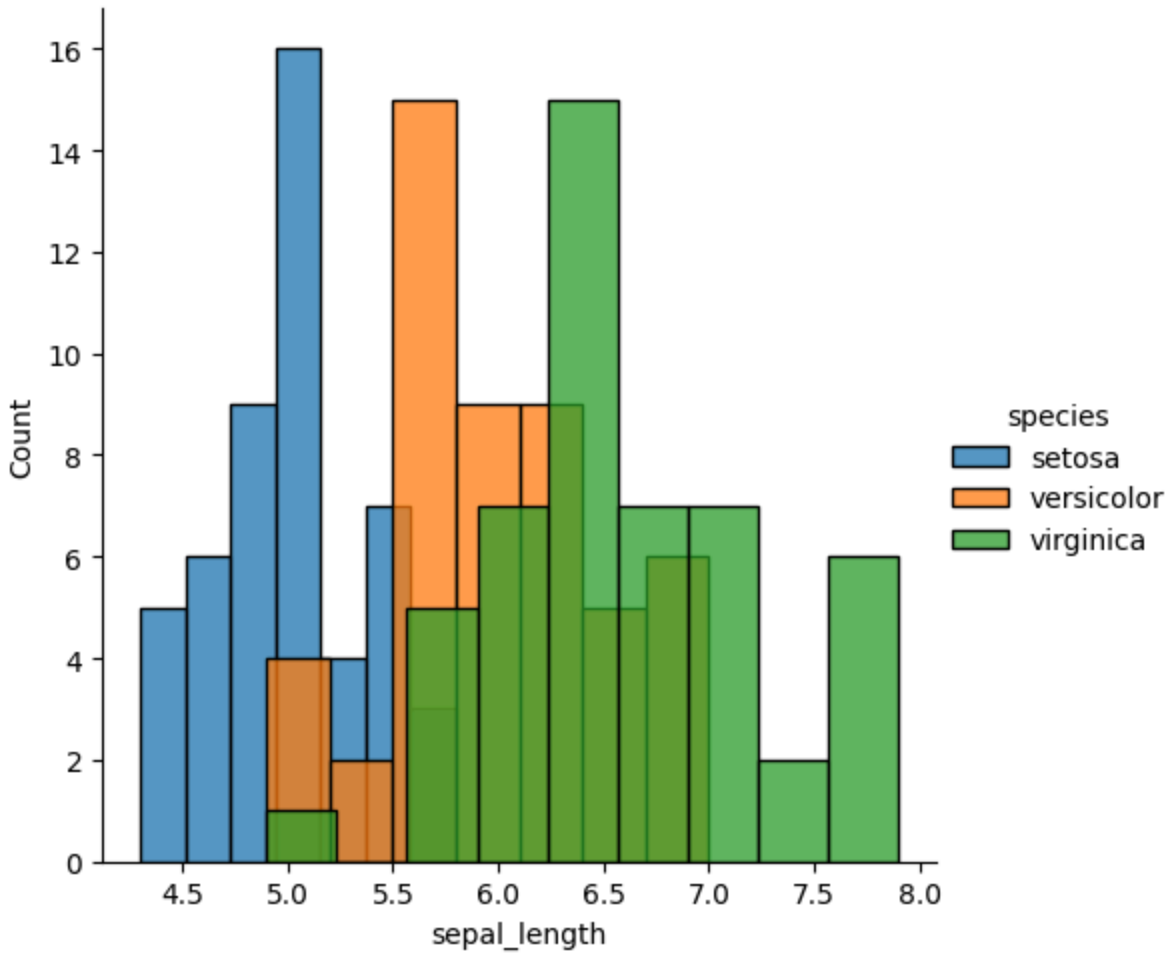
<Axes: xlabel='petal_length', ylabel='petal_width'>

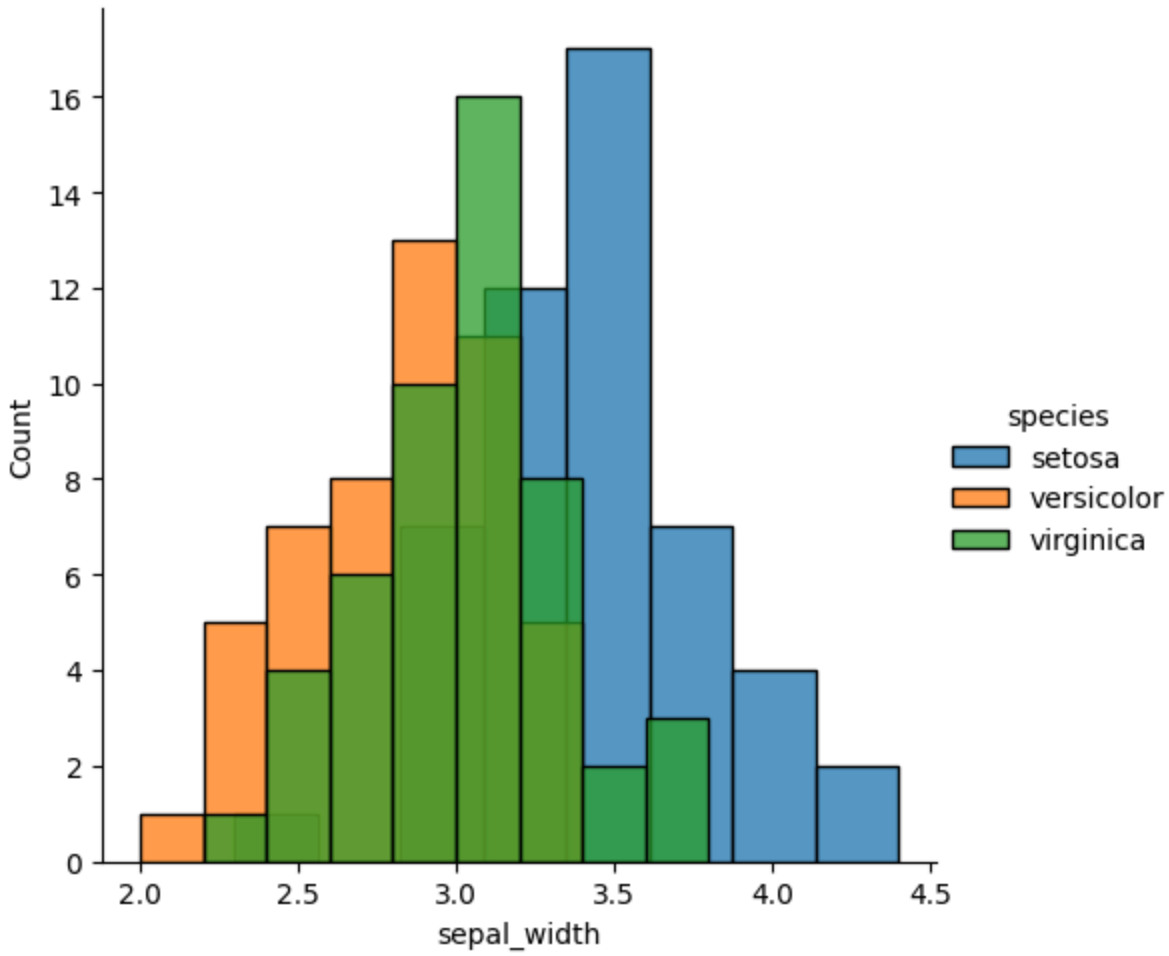

species
setosa
versicolor
virginica
petal_width
petal_length
0.0
0.5
1.0
1.5
2.0
2.5
1
2
3
4
5
6
7


sepal_length
sepal_width
petal_length
petal_width
species
setosa
versicolor
virginica


Count
petal_length
species
setosa
versicolor
virginica
0
2
4
6
8
10
12
14
1
2
3
4
5
6
7


Count
25
20
15
10
5
0
0.0
0.5
1.0
1.5
2.0
2.5
petal_width
species
setosa
versicolor
virginica


16
14
12
10
8
6
4
2
0
Count
4.5
5.0
5.5
6.0
6.5
7.0
7.5
8.0
sepal_length
species
setosa
versicolor
virginica



RESULT:

The experiment has been carried out successfully